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Numerical Technique Laboratory

Project Proposal

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Project Title:	Continuous-Time System Response Simulation of a Biomedical Signal Filter using Numerical ODE Solvers

1 Abstract

This project aims to simulate the response of a continuous-time second-order Butterworth high-pass filter, implemented as an analog circuit, to a corrupted ECG signal containing baseline wander noise. The filter is modeled as a continuous-time Linear Time-Invariant (LTI) system governed by a second-order ordinary differential equation (ODE), which is converted to state-variable form. Instead of relying on built-in MATLAB solvers, we will implement a custom 4th-order Runge-Kutta (RK4) time-stepping loop and central finite difference for numerical differentiation to simulate the system's zero-state response step-by-step. The project will evaluate the filter's effectiveness in removing low-frequency drift, analyze the impact of numerical methods on accuracy, and provide time-domain visualizations of the filtering process.

2 Background

Electrocardiogram (ECG) monitoring is a critical tool in modern healthcare for diagnosing cardiac conditions. However, the clinical utility of raw ECG recordings is often heavily corrupted by various types of noise and artifacts. Among these, **baseline wander**—a low-frequency noise artifact—is particularly problematic as it distorts diagnostic features such as the ST segment and other low-frequency components essential for accurate diagnosis.

Baseline wander typically arises from patient respiration, body movement, and poor skin-electrode contact. The frequency content of this noise is generally below 0.5 Hz, though increased physical activity during stress tests can elevate this range, typically lying between 0.1 Hz and 1 Hz. This uncorrected low-frequency interference can mask or mimic subtle but clinically significant features such as ST-segment changes, which are critical for diagnosing myocardial ischemia, thereby significantly hurting diagnostic reliability.

To extract true clinical data, the signal must be processed through a high-pass filter to remove the unwanted low-frequency drift. Traditional approaches often employ digital IIR Butterworth filters with cutoff frequencies around 0.5 Hz to 0.75 Hz. However, these methods typically use built-in, black-box functions that obscure the underlying numerical principles. This project takes a pedagogical approach by explicitly treating the filter as a physical continuous-time LTI system and solving its governing ODE numerically. By focusing on time-domain system representations and zero-state responses, we connect circuit theory directly with the RK4 and finite-difference techniques from the numerical methods bridging theoretical signals and systems analysis with practical biomedical engineering applications.

3 Complexity Analysis

This project demonstrates several Complex Engineering Attributes:

- i. **Mathematical Modeling and Analysis:** The project requires formulating a second-order linear ODE that models a physical analog high-pass filter circuit. Converting this ODE into a state-space representation involving two coupled first-order ODEs demonstrates the ability to abstract a physical system into a mathematical framework suitable for computational analysis.
- ii. **Numerical Methods and Algorithm Design:** Implementing a custom 4th-order Runge-Kutta (RK4) time-stepping loop involves designing a stable and accurate numerical integration scheme. Applying the central finite difference method for numerical differentiation requires careful consideration of step size selection and error analysis, bypassing standard "plug-and-chug" toolbox functions.
- iii. **System Analysis in Time and Frequency Domains :** The project involves analyzing the filter's behavior through multiple output visualizations, including time-domain comparisons of input vs. output signals, error/residue plots, and parametric analysis of numerical methods. This multi-perspective analysis mirrors real-world engineering problem-solving.
- iv. **Engineering Trade-offs and Parametric Analysis:** A key component is investigating how changing the numerical step size (Δt) or selecting a simpler method (like Euler's method) affects the stability and computational accuracy. This balances filter cutoff accuracy against numerical stability and computational cost—a classic engineering trade-off where choices carry real consequences for patient safety via signal fidelity.
- v. **Integration of Theory and Practice:** The project bridges theoretical concepts from signals and systems (LTI systems, state-space) with practical implementation using MATLAB, demonstrating the application of theoretical knowledge to a real-world biomedical signal processing problem.

4 Technical Requirements

The project will be implemented using MATLAB and will require the following:

- i. **Software Environment:**
 - MATLAB R2020a or later.
 - Basic MATLAB functionality (no specialized toolboxes required as the RK4 is hand-coded).
 - Signal Processing Toolbox (optional, solely for cross-checking validation).
- ii. **Signal Generation:**
 - Synthetic clean ECG signal constructed using combinations of Gaussian curves or triangular pulses.
 - Synthetic noise signal: sinusoidal low-frequency drift at 0.2 Hz.
 - MATLAB's built-in awgn function for optional additive noise testing.

iii. **Numerical Implementation:**

- Custom RK4 time-stepping loop.
- Central finite difference method for second derivative calculation.
- Configurable time step (Δt) and simulation duration.

iv. **Filter Parameters:**

- Second-order Butterworth high-pass filter characteristics: Cutoff frequency $f_c = 0.5$ Hz ($\omega_n = 2\pi \times 0.5$ rad/s), Damping ratio $\zeta = 0.707$.
- State variables: $z_1 = y$ (output), $z_2 = dy/dt$ (derivative of output).
- Initial conditions: $z_1(0) = 0$, $z_2(0) = 0$ (zero-state response).

v. **Output Visualizations:**

- Basic plotting functions for time-domain comparison ($x(t)$ vs. $y(t)$), error/residue plots, and parametric comparisons.

5 Proposed Methodology

1. Mathematical Modeling

The filter is modeled as a second-order Butterworth high-pass filter with the governing differential equation:

$$\frac{d^2 y}{dt^2} + 2\zeta\omega_n \frac{dy}{dt} + \omega_n^2 y = \frac{d^2 x}{dt^2}$$

where $x(t)$ is the corrupted input signal, $y(t)$ is the clean output signal, $\omega_n = 2\pi \times 0.5$ rad/s, and $\zeta = 0.707$. We convert this into a state-space representation by letting $z_1 = y$ and $z_2 = dy/dt$, yielding a coupled system:

$$\begin{aligned} \frac{dz_1}{dt} &= z_2 \\ \frac{dz_2}{dt} &= -2\zeta\omega_n z_2 - \omega_n^2 z_1 + \frac{d^2 x}{dt^2} \end{aligned}$$

The input signal's second derivative is computed using the central finite difference method:

$$\frac{d^2 x}{dt^2} \approx \frac{x(t + \Delta t) - 2x(t) + x(t - \Delta t)}{(\Delta t)^2}$$

2. Synthetic Signal Design

Generate a clean ECG trace using a combination of localized Gaussian curves or rapid triangular pulses to simulate the PQRST complex morphology at approximately 72 BPM. Baseline wander noise will be modeled as:

$$x_{noise}(t) = A \cdot \sin(2\pi \cdot 0.2 \cdot t)$$

where A is an amplitude factor (approx. 10-20% of the ECG amplitude). The corrupted input signal

is then:

$$x(t) = x_{ECG}(t) + x_{noise}(t)$$

3. Numerical Implementation

The 4th-order Runge-Kutta (RK4) method will be implemented as an explicit time-stepping loop. For each time step i , calculate four slope coefficients for both state variables:

For state z_1 :

$$\begin{aligned} k_{1,1} &= f_1(t_i, z_{1,i}, z_{2,i}) = z_{2,i} \\ k_{1,2} &= f_1\left(t_i + \frac{\Delta t}{2}, z_{1,i} + \frac{\Delta t}{2} k_{1,1}, z_{2,i} + \frac{\Delta t}{2} k_{2,1}\right) \\ k_{1,3} &= f_1\left(t_i + \frac{\Delta t}{2}, z_{1,i} + \frac{\Delta t}{2} k_{1,2}, z_{2,i} + \frac{\Delta t}{2} k_{2,2}\right) \\ k_{1,4} &= f_1(t_i + \Delta t, z_{1,i} + \Delta t \cdot k_{1,3}, z_{2,i} + \Delta t \cdot k_{2,3}) \end{aligned}$$

For state z_2 :

$$\begin{aligned} k_{2,1} &= f_2(t_i, z_{1,i}, z_{2,i}) = -2\zeta\omega_n z_{2,i} - \omega_n^2 z_{1,i} + \frac{d^2x}{dt^2}(t_i) \\ k_{2,2} &= f_2\left(t_i + \frac{\Delta t}{2}, z_{1,i} + \frac{\Delta t}{2} k_{1,1}, z_{2,i} + \frac{\Delta t}{2} k_{2,1}\right) \\ k_{2,3} &= f_2\left(t_i + \frac{\Delta t}{2}, z_{1,i} + \frac{\Delta t}{2} k_{1,2}, z_{2,i} + \frac{\Delta t}{2} k_{2,2}\right) \\ k_{2,4} &= f_2(t_i + \Delta t, z_{1,i} + \Delta t \cdot k_{1,3}, z_{2,i} + \Delta t \cdot k_{2,3}) \end{aligned}$$

The state variables are then updated as:

$$\begin{aligned} z_{1,i+1} &= z_{1,i} + \frac{\Delta t}{6} (k_{1,1} + 2k_{1,2} + 2k_{1,3} + k_{1,4}) \\ z_{2,i+1} &= z_{2,i} + \frac{\Delta t}{6} (k_{2,1} + 2k_{2,2} + 2k_{2,3} + k_{2,4}) \end{aligned}$$

4. Parametric Analysis

To demonstrate the impact of numerical method choice on accuracy and stability, the simulation will be repeated under various conditions: varying step sizes ($\Delta t = 0.01, 0.005, 0.001$ seconds), comparing RK4 vs. Euler's method, and evaluating Mean Squared Error, PSNR, and computation time.

5. Expected Outputs and Plots

1. **Time-Domain Comparison Plot:** Showing the heavily drifting raw input signal $x(t)$ alongside the stable, centered output signal $y(t)$.
2. **Error and Residue Plot:** Highlighting the isolated baseline wander profile extracted by the numerical filter loop to track the mathematical noise model.

3. **Parametric Analysis Plot:** A comparative view showing how Δt variations or Euler's method affects stability and accuracy.

6 References

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